Build and Run a Java Application with Maven, Migrate the Same Application to Gradle.

Procedure:

1. Create a **Maven project** by specifying the *groupId* and *artifactId*.
2. Under `src/main/java`, create a package and write a Java class with the required program logic (include `main()` method if execution is needed).
3. Under `src/test/java`, create a package and write test classes for the application.
4. Add the required **TestNG dependency** in the `pom.xml` file.
5. Open the **Terminal** in the IDE
   (Window → Show View → Terminal).
6. Compile the project using Maven command:
   `mvn clean compile`
7. Run the Java program (if `main()` is present) using:
   `java -cp target/classes packageName.className`
8. Execute test cases using Maven command:
   `mvn test`
9. Verify test execution reports in the following location:
   `target/surefire-reports/`
10. Install **Gradle**, initialize it in the same project using `gradle init`, and build the application using Gradle.

Code

src/main/java:

```
package pgm1;
public class Result
{
	public static String display(int m)
	{
		if(m<0 || m>100)
			return "invalid";
		else if(m<40)
			return "fail";
		else
			return "pass";
	}
	public static void main(String[] args)
	 {
		System.out.println(display(55));
	 }
}
```

src/test/java:

```
package pgm1;

import org.testng.Assert;
```

```
import org.testng.annotations.Test;

public class ResultTest {

    @Test
    public void testPass() {
        Assert.assertEquals(Result.display(55), "pass");
    }

    @Test
    public void testFail() {
        Assert.assertEquals(Result.display(35), "fail");
    }

    @Test
    public void testInvalid() {
        Assert.assertEquals(Result.display(-5), "invalid");
    }
}
```

In **pom.xml** add the following dependency to run Testcases.

```
<dependencies>
  <!-- https://mvnrepository.com/artifact/org.testng/testng -->
<dependency>
    <groupId>org.testng</groupId>
    <artifactId>testng</artifactId>
    <version>7.5.1</version>
    <scope>test</scope>
</dependency>
</dependencies>
```

Output:

```
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.608 s
[INFO] Finished at: 2026-01-25T17:32:36+05:30
```